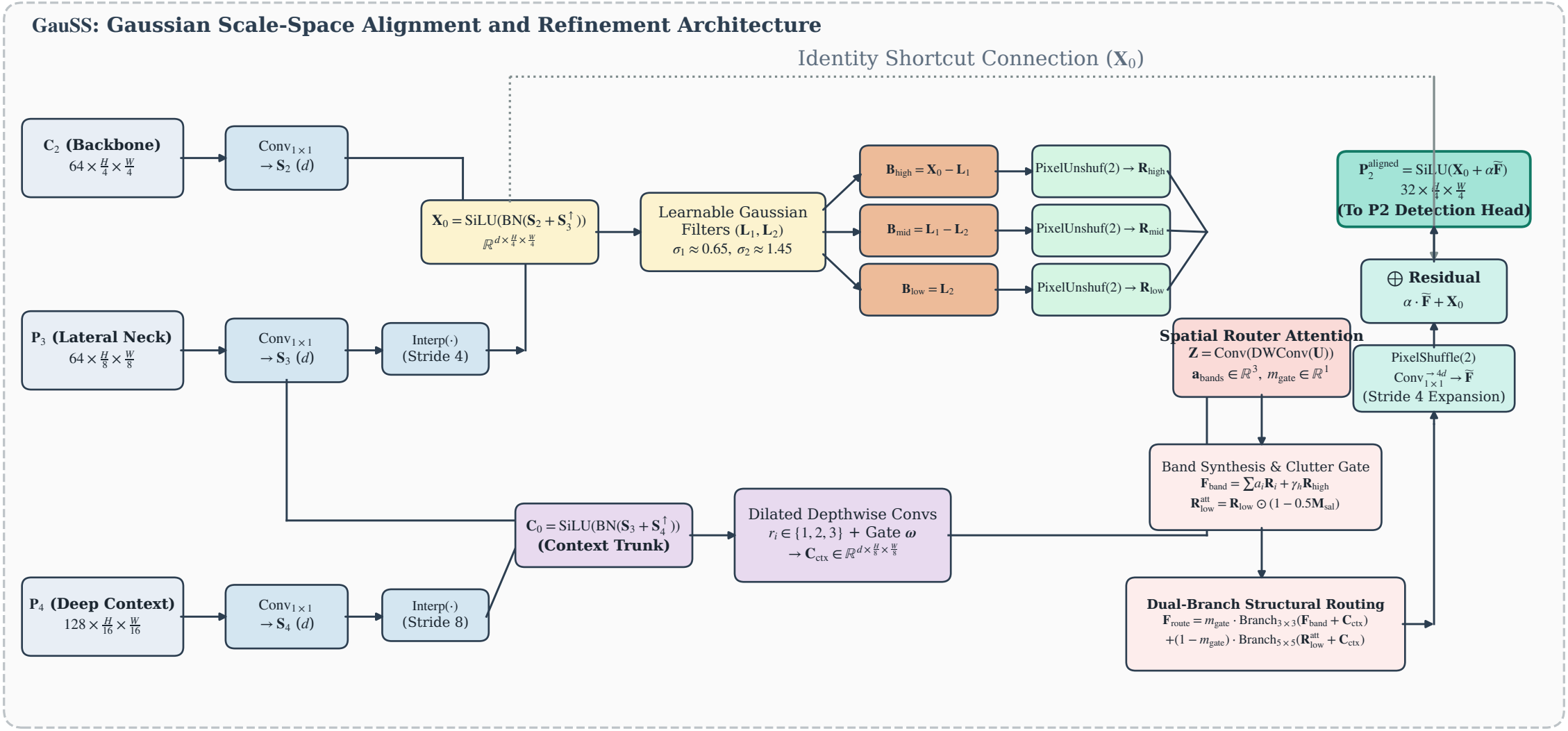
GauSS: Gaussian Scale-Space Alignment and Refinement Architecture
Identity Shortcut Connection ($\mathbf{X}_0$)
$\mathbf{C}_2$ (Backbone)
$64 \times \frac{H}{4} \times \frac{W}{4}$
$\text{Conv}_{1\times1} \rightarrow \mathbf{S}_2\ (d)$
$\mathbf{X}_0 = \text{SiLU}(\text{BN}(\mathbf{S}_2 + \mathbf{S}_3^{\uparrow}))$
$\mathbb{R}^{d \times \frac{H}{4} \times \frac{W}{4}}$
Learnable Gaussian Filters ($\mathbf{L}_1, \mathbf{L}_2$)
$\sigma_1 \approx 0.65,\ \sigma_2 \approx 1.45$
$\mathbf{B}_{\text{high}} = \mathbf{X}_0 - \mathbf{L}_1$
$\mathbf{B}_{\text{mid}} = \mathbf{L}_1 - \mathbf{L}_2$
$\mathbf{B}_{\text{low}} = \mathbf{L}_2$
PixelUnshuf(2) $\rightarrow \mathbf{R}_{\text{high}}$
PixelUnshuf(2) $\rightarrow \mathbf{R}_{\text{mid}}$
PixelUnshuf(2) $\rightarrow \mathbf{R}_{\text{low}}$
$\mathbf{P}_3$ (Lateral Neck)
$64 \times \frac{H}{8} \times \frac{W}{8}$
$\text{Conv}_{1\times1} \rightarrow \mathbf{S}_3\ (d)$
Interp(·) (Stride 4)
$\mathbf{P}_4$ (Deep Context)
$128 \times \frac{H}{16} \times \frac{W}{16}$
$\text{Conv}_{1\times1} \rightarrow \mathbf{S}_4\ (d)$
Interp(·) (Stride 8)
$\mathbf{C}_0 = \text{SiLU}(\text{BN}(\mathbf{S}_3 + \mathbf{S}_4^{\uparrow}))$
(Context Trunk)
Dilated Depthwise Convs
$r_i \in \{1, 2, 3\}$ + Gate $\boldsymbol{\omega}$
$\rightarrow \mathbf{C}_{\text{ctx}} \in \mathbb{R}^{d \times \frac{H}{8} \times \frac{W}{8}}$
Spatial Router Attention
$\mathbf{Z} = \text{Conv}(\text{DWConv}(\mathbf{U}))$
$\mathbf{a}_{\text{bands}} \in \mathbb{R}^3,\ m_{\text{gate}} \in \mathbb{R}^1$
Band Synthesis & Clutter Gate
$\mathbf{F}_{\text{band}} = \sum a_i \mathbf{R}_i + \gamma_h \mathbf{R}_{\text{high}}$
$\mathbf{R}_{\text{low}}^{\text{att}} = \mathbf{R}_{\text{low}} \odot (1 - 0.5\mathbf{M}_{\text{sal}})$
Dual-Branch Structural Routing
$\mathbf{F}_{\text{route}} = m_{\text{gate}} \cdot \text{Branch}_{3\times3}(\mathbf{F}_{\text{band}} + \mathbf{C}_{\text{ctx}})$
$+(1 - m_{\text{gate}}) \cdot \text{Branch}_{5\times5}(\mathbf{R}_{\text{low}}^{\text{att}} + \mathbf{C}_{\text{ctx}})$
PixelShuffle(2)
$\text{Conv}_{1\times1}^{\rightarrow 4d} \rightarrow \widetilde{\mathbf{F}}$
(Stride 4 Expansion)
$\oplus$ Residual
$\alpha \cdot \widetilde{\mathbf{F}} + \mathbf{X}_0$
$\mathbf{P}_2^{\text{aligned}} = \text{SiLU}(\mathbf{X}_0 + \alpha\widetilde{\mathbf{F}})$
$32 \times \frac{H}{4} \times \frac{W}{4}$
(To P2 Detection Head)